



Photonfocus AG

Photonfocus SDK for Linux

Quickstart guide

1 PFSDK for Linux quickstart guide

After extracting the tarball (in which this guide is included) the following directory structure can be found.

- *bin*: contains any executables packaged with the PFSDK.
- *doc*: documentation for the different APIs.
- *driver*: network filter driver for the Linux Kernel.
- *examples*: sample projects showing SDK usage.
- *include*: library header files.
- *lib*: library object files.

The first thing the user should do is decide whether to install the filter driver. Installing the filter driver will provide better performance. Whether the filter driver is used or not can be configured in the API.

1.1 Installing the filter driver

Kernel modules need to be recompiled for each minor version change in the kernel. Hence, providing a driver binary for all kernel versions would not be feasible. For convenience, a DKMS install is provided, which will install the driver source and recompile it every time there is a kernel change.

1.1.1 DKMS install

This requires the system to have DKMS installed. In an Ubuntu distribution, DKMS may be installed with the following command:

```
sudo apt-get install dkms
```

After installing DKSM, execute the script found in the *dkms* folder with root privileges.

```
sudo ./install_with_dkms.sh
```

1.2 Building and running the samples

The samples are provided with a CMake configuration file. Install CMake 3.12 or greater before proceeding. The following pattern can be used to configure, build, and run the samples:

```
cd examples/C++/DiscoverConfigAndGrab_Console
mkdir build
cd build
cmake ..
make
./PFCameraLib_DiscoverConfigAndGrab_Console
```

If the *examples* folder is moved in relation to the *lib* folder and the SDK was not extracted into one of the standard CMake search paths, additional environment variables may need to be specified.

Visit the following link for more information about CMake's `find_package`:

https://cmake.org/cmake/help/latest/command/find_package.html#search-procedure

1.3 Read the documentation

The documentation is provided in HTML format. Use a browser to navigate the documentation:

- <doc/PFCameraLib/html/index.html> (C++ API)
- <doc/PFPyCameraLib/index.html> (Python API)